



2 Real Analysis 2023

Tutorial 4

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Limits of sequences

Problem 13 Let $(a_n)_{n=1}^{\infty}$ be a sequence with $\lim_{n \rightarrow \infty} a_n = 4$. Prove by definition that

$$(a) \lim_{n \rightarrow \infty} (3a_n - 2) = 10, \quad (b) \lim_{n \rightarrow \infty} a_n(a_n + 1) = 20, \quad (c) \lim_{n \rightarrow \infty} \frac{1}{a_n} = \frac{1}{4}.$$

Problem 14 Prove by definition that $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$.

Problem 15 Use the Squeeze Theorem and rules for working with limits to calculate the following limits:

$$(a) \lim_{n \rightarrow \infty} \frac{n \cos(n^2)}{2n^2 + 1}, \quad (b) \lim_{n \rightarrow \infty} \frac{1 + (-1)^n}{n}.$$

Problem 16 Recall that a sequence $(a_n)_{n=1}^{\infty}$ is called *eventually constant* if there are $a \in \mathbb{R}$ and $N \in \mathbb{N}$ such that $a_n = a$ for all $n \geq N$.

Prove or disprove:

- (a) Every eventually constant sequence is convergent.
- (b) Every convergent sequence is eventually constant.
- (c) Every eventually constant sequence is bounded.